

27. Farzana refinances her loans for a reduced interest rate before she begins to pay them off. The total of her refinanced loan is \$39,447.50. The new interest rate is 6.99%. How much are Farzana's payments if she has a 10-year term on this loan?
28. Carly refinances her loans for a reduced interest rate before she begins to pay them off. The total of her refinanced loan is \$61,332.00. The new interest rate is 7.45%. How much are Carly's payments if she has a 10-year term on this loan?
29. Andrew consolidates his federal student loans to make paying them easier. They total \$27,582.00. The consolidated loan is at 4.99% interest. How much are Andrew's payments if his new term is 15 years?
30. Manuel consolidates his federal student loans to make paying them easier. They total \$19,470.00. The consolidated loan is at 4.99% interest. How much are Manuel's payments if his term is 10 years?
31. Steven has four members in his Georgia household and makes \$53,400 annually. The federal poverty guideline for him is \$27,750. What is Steven's discretionary income?
32. Akash has three members in his New Jersey household and makes \$38,120 annually. The federal poverty guideline for him is \$23,030. What is Akash's discretionary income?
33. Kateryna has two members in her Iowa household and earns \$42,500 annually. The federal poverty guideline for her is \$18,310. She qualifies for a REPAYE repayment plan. What is the maximum monthly payment Kateryna would pay?
34. Keirstin lives by herself in her New Mexico home and earns \$28,530 annually. The federal poverty guideline for her is \$13,590. She qualifies for a REPAYE repayment plan. What is the maximum monthly payment Keirstin would pay?

6.10 Credit Cards



Figure 6.22 Credit cards are both a convenience and a danger. (credit: "Credit Cards" by Sean MacEntee/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Apply for a credit card armed with basic knowledge.
2. Distinguish between three basic types of credit cards.
3. Compare and contrast the benefits and drawbacks of credit cards.
4. Read and understand the basic parts of a credit card statement.
5. Compute interest, balance due, and minimum payment due for a credit card.

It can be difficult to get along these days without at least one credit card. Most hotels and rental car agencies require that a credit card is used. There are even a number of retailers and restaurants that no longer accept cash. They make online purchasing easier. And nothing contributes more to a good credit rating than a solid history of making credit card payments on time.

Being granted a credit card is a privilege. Used unwisely that privilege can become a curse and the privilege may be withdrawn. In this section, we will talk about the different types of credit cards and their advantages and disadvantages.

The more knowledge a cardholder has about the credit card industry, the better able credit accounts can be managed, and that knowledge may cause major adjustments to a cardholder's lifestyle.

All credit cards are not equal, but they all represent consumers borrowing money, usually from a bank, to pay for needs and “wants.” As such, they are a type of loan, and your repayment may include interest. (You might want to review Section 6.8, which discusses loans and repayment plans.)

There are many institutions and credit cards to choose from. Use caution as you shop around for a credit card that suits you. Your top concern is likely the interest rates on purchases and cash advances. But be careful to also read the small print regarding charges for late payments, and other fees such as an **annual fee**, where the credit card charges you (the cardholder) a fee each year for the privilege of using the cards. Many cards charge no such fee, but there are many that charge modest to heavy fees. Make sure to understand rules for **reward programs**, where the credit card issuer grants benefits based on one's spending. Finally, once one applies for and is granted a credit card, pay attention to the **credit limit** the bank offers. Once a company is owed that much money, use of the card for purchases should be curtailed until some of the debt is paid off.

 *The interest rate will not matter if the balance is paid every month. When the balance is paid every month, there is NO INTEREST charged.*

Types of Credit Cards

There are basically three types of credit cards: bank-issued credit cards, store-issued credit cards, and travel/entertainment credit cards. We will look at all three and explain the good and the bad qualities of each.

Bank-Issued Credit Cards

Perhaps the most widely used credit card type is the **bank-issued credit card**, like Visa or MasterCard (and even American Express and Discover cards). These types of cards are an example of revolving credit, meaning that additional credit is extended before the previous balance is paid—but only up to the assigned credit limit. Bank-issued cards are considered the most convenient, as they can be used to purchase anything, including apparel, furniture, groceries, fuel for automobiles, meals, hotel bills, and so on, just as if paying with cash. The interest rates on bank-issued credit cards are usually lower than those for other credit cards we'll discuss, and the credit limits are generally higher. Currently, bank-issued cards have an average 20.09% APR.

Store-Issued Credit Cards

Store-issued credit cards are issued by retailers. One can hardly walk into a store these days without being offered a discount on purchases if one applies for the store credit card. These cards can only be used in that store or family of stores that issues the card. However, if a store credit card is associated with Visa, MasterCard, or American Express, then the card might be used the same way that the bank-issued cards are used. This is called cobranding. The logo of the bank-issued card will be present on the store card. Many stores offer both types. Like other credit cards, they may come with an annual fee.

Store credit cards usually charge higher interest rates than bank-issued cards. Currently, store credit cards have an APR (annual percentage rate) of 24.15%. Any rewards offered by store credit cards are usually limited to purchases made in their own store, and it typically takes longer to accumulate enough rewards or points to redeem them, whereas cobranding credit cards offer opportunities to earn rewards on all purchases, regardless of whether purchases are made in the issuing store or not.

Store credit cards usually offer lower credit limits, at least in the beginning. After being proved to be a responsible credit card owner, credit limits can be raised. Nevertheless, store credit cards are a good choice for those new to the credit card industry. If on-time payments are consistently made, it is an excellent way to get started building a credit history.

Travel/Entertainment Cards, or Charge Cards

This is the third type of credit card. The **travel and entertainment cards**, also known as **charge cards**, first and foremost offer very high limits or unlimited credit, but they must be paid in full every month. They generally charge high annual fees and impose expensive penalties should a payment be late. On the other hand, they typically have longer grace periods and offer many and various kinds of rewards.

Check out this [nerdwallet article about the differences between a charge card and a credit card \(https://openstax.org/r/cards_charge_cards\)](https://openstax.org/r/cards_charge_cards).

 *An interest rate will greatly depend on credit score. Responsible use of credit cards will increase a credit score. See*

the WHO KNEW? from [The Basics of Loans](#).

▶ VIDEO

[Choosing a Credit Card \(https://openstax.org/r/Credit_Card\)](https://openstax.org/r/Credit_Card)

?? WHO KNEW?

Top Travel and Entertainment Cards

At one time, Diner's Club was the premier entertainment card. To be accepted into the Diner's Club and be rewarded with a charge card meant one was special. Today, an American Express card is held with the same reverence as Diner's Club was in the past. It was a privilege to own one of these cards. There is a lot of competition going on to supplant Diner's Club, as shown by the [Chase Sapphire Preferred Card's \(https://openstax.org/r/Chase_Sapphire\)](https://openstax.org/r/Chase_Sapphire) travel rewards and benefits.

Another thing worth mentioning here and something that appears to be an unusual offering is part of the American Express Gold and higher-level cards. Cardholders can actually open a savings account, buy a CD, or apply for a personal or business loan from American Express. The company boasts a higher interest rate than what is paid on traditional savings accounts and CDs.

EXAMPLE 6.91

Comparing Credit Cards

1. Which type of credit card is paid off every month so has no interest to be paid, but comes with high fees?
2. Which type of credit cards are the most widely accepted?
3. Which type of credit cards are the most limited?

✓ Solution

1. Charge cards are to be paid off completely each month.
2. Bank-issued credit cards are the most flexible to use, because they are not limited to which retailers or service providers accept them.
3. Store issues credit cards are the most limited, since they only work in that family of stores.

> YOUR TURN 6.91

1. Which type of credit card typically have the highest interest rates?
2. What should be considered other than interest rate when selecting a card?
3. Which type of cards typically have high annual fees?

Credit Card Statements

Cardholders usually receive monthly statements and have 21 days to pay the minimum amount due. The statements itemize and summarize activity on the credit card for that statement's **billing period**. The billing period for a credit card is generally a month long, but typically does not start and end on the first and last days of the month. The statement will include the current balance, interest rate, the minimum payment due, and the due date. Be aware, different companies produce statements that are laid out differently. The information will be clearly labeled though.

The due date is a top concern. Missing a due date is one of the worst things a cardholder can do financially, and this is by far the biggest downfall of owning a credit card. Not only is the cardholder subject to late fees, but when a payment is late more than once there is a high probability that the cardholder will be negatively reported to the credit bureaus, which can quickly erode a credit score. [Figure 6.23](#) shows an excerpt from an actual statement from a Chase Bank Visa card, based on the current \$668.25 balance.

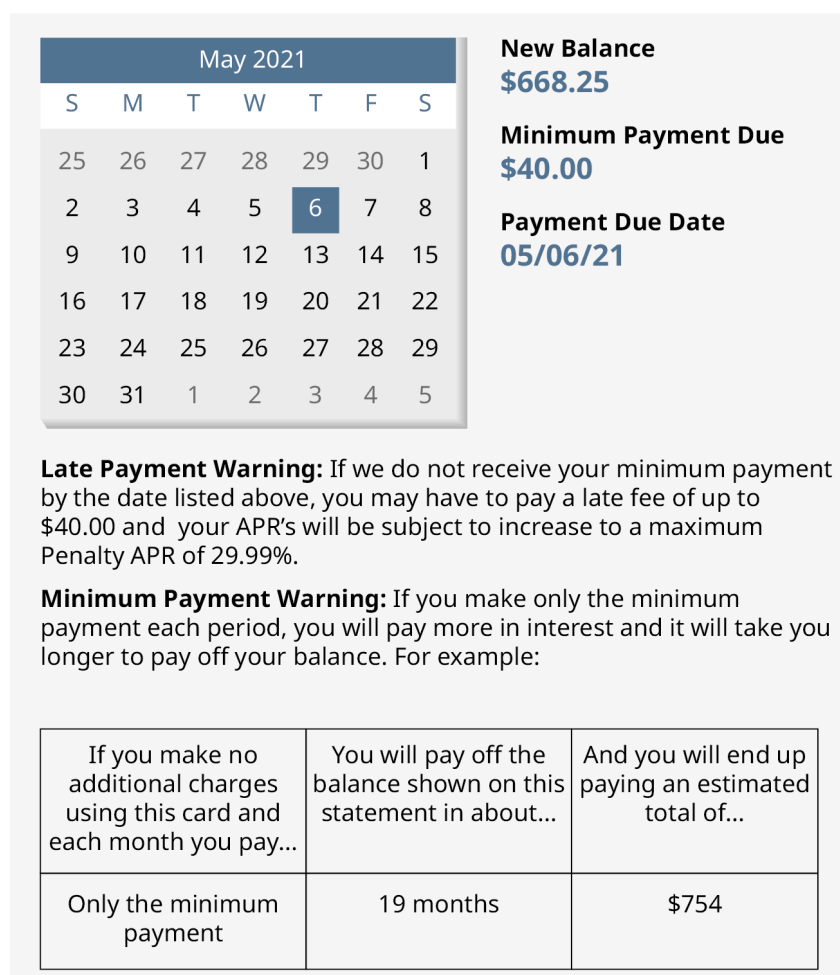


Figure 6.23 Credit card statement

Specifically pay attention to the late payment penalty and minimum payment warning statements. Stating that if no other purchases are made and you continue making only the minimum payment, it will take 19 months to pay off the balance and you will pay \$754.00. You can't say you were not warned.

It is critical that you examine your statement every month because it is always a possibility that your account may have been compromised. If you should notice fraudulent charges on your statement, notifying the credit card company is often enough to have those charges researched by the company and removed. The card with the fraudulent charges will be canceled and a new card with a new account number will be sent to you.

EXAMPLE 6.92

Reading a Credit Card Statement

On the credit card statement [Figure 6.24](#), identify

1. The balance due
2. The minimum required payment
3. The length of time it takes to pay off the balance by paying the minimum payments and without charging more to the card
4. The interest rate for purchases


YourBank

Your Credit Card Account Statement

PAYMENT INFORMATION

New Balance:	\$ 3663.23
Your Minimum Payment:	\$ 36.63
Your Minimum Payment Due Date:	May 24, 2015

Estimated time to pay
The estimated time to pay your New Balance in full if you pay only the Minimum Payment each month is 2 year(s) and 4 month(s).

SUMMARY OF YOUR ACCOUNT

Previous Statement Balance:	\$ 2654.48
Payments:	\$-2654.48
New Purchases:	\$ 1957.24
Balance Transfers and Access Cheques:	\$ 1200.00
Cash Advances	\$ 500.00
Interest:	\$ 0.00
Fees:	\$ 5.99
Subtotal	\$ 3663.23
Your New Balance:	\$ 3663.23
Credit Limit:	\$ 9000.00
Cash Advance Limit:	\$ 500.00
Credit Available:	\$ 5336.77
Cash Advance Limit:	\$ 0.00
Statement Closing Date:	April 30, 2015
Days in Statement Period:	30
Annual Interest Rate for Purchases:	19.99%
Annual Interest Rate for Balance Transf. and Access Cheques:	2.50%
Annual Interest Rate for Cash Advances:	19.99%

Statement Period:
 From April 01, 2015
 To April 30, 2015

 Account Number:
 9999 99XX XXXX 1234

 Primary Cardholder:
 Yourname Yoursurname

Ways to pay:
 Online Banking
 Telephone Banking
 ATM
 Pie-Authorized Payment
 By Mail To:
 YourBank
 P.O. Box 1234 Section Z
 BankCity, NY, 98456

Contact Information:
www.websitename.com
 Customer Service/Lost or Stolen
 1-888-123-4567
 TTY/TDD
 1-888-123-4567


YourBank

Payment Slip

PAYEE FIRST NAME LAST NAME
 PAYEE ADDRESS:
 99 STREET NAME
 CITY NAME, ZIP CODE
 STATE, COUNTRY

Account Number: **9999 99XX XXXX 123**
 Your New Balance: **\$ 3663.23**
 Your Minimum Payment: **\$ 36.63**
 Your Minimum Payment Due Date: **May 24, 2015**

 Amount you're paying: **\$ 3663.23**

999 999 999 99999 999 9999 99XX XXXX 1234 XX

Figure 6.24 Credit card account statement

✓ **Solution**

1. The balance due is under the payment information heading and is \$3,663.23.
2. The minimum payment due is also under the payment information heading, and is \$36.63.
3. The time to pay off the balance using only minimum payments is below the payment information, and says it takes 2 years and 4 months to pay off the balance.
4. The interest rate for purchases is toward the bottom of the statement. It is 19.99%.

> YOUR TURN 6.92

Referring to the statement above, answer the following:

1. What is the statement period?
2. What is the credit limit?
3. How much in fees were charged?

▶ VIDEO

[Reading Credit Card Statements \(https://openstax.org/r/Reading_Credit\)](https://openstax.org/r/Reading_Credit)

Compute Interest, Balance Due, and Minimum Payment Due for a Credit Card

Computing all of these values depends on understanding and computing the average daily balance on a credit card. Once that is known, the interest, balance due, and minimum payment can be found.

Above all else, if you pay off the entire balance each month, interest is not charged.

Average Daily Balance

Most credit card companies compute interest using the **average daily balance** method.

To find the average daily balance on your credit card, determine the balance on the card each day of the billing period (often that month), and take the average. One process to find that average daily balance follows these steps:

1. Start with a list of transactions with their dates and amounts.
2. For each day that had transactions, find the total of the transactions for the day. Expenditures are treated as positive values, payments are treated as negative values.
3. Create a table containing each day with a different balance. The balance is the previous balance plus the day's total transactions.
4. Add a column for the number of days those balances until the balance changed.
5. Add a column that contains the balances multiplied by the number of days until the balance changed.
6. Find the sum of that last column.
7. Divide the sum by the number of days in the billing period (often the number of days in the month). This is the average daily balance.

EXAMPLE 6.93

Computing Average Daily Balance

The billing cycle goes from May 1 to May 31. The balance at the start of the billing cycle is \$450.21. The list of transactions on the card is below.

Date	Activity	Amount
1-May	Billing Date Balance	\$450.21
10-May	Payment	\$120.00
15-May	Groceries	\$83.43
26-May	Auto Parts	\$45.12
26-May	Restaurant	\$85.34
30-May	Shoes	\$98.23

Find the average daily balance for the credit card during the month of May.

✓ **Solution**

To find the average daily balance, we use the following steps.

1. Start with a list of transactions with their dates and amounts.
This list is provided.
2. For each day that had transactions, find the total of the transactions for the day.
The only day with more than one transaction was May 26. The sum of those transactions is \$130.46. Treating the payment as a negative value, the daily transaction amounts are

Date	Amount
1-May	\$450.21
10-May	-\$120.00
15-May	\$83.43
26-May	\$130.46
30-May	\$98.23

3. Create a table containing each day with a different balance.
The new table with dates that had different balances is below.

Date	Balance
1-May	\$450.21
10-May	\$330.21
15-May	\$413.64
26-May	\$544.10
30-May	\$642.33

4. Now, add a column for the number of days those balances until the balance changed. The days until the balance changes is found by finding the difference in the dates. For instance, from May 15 to May 26 was 11. Adding that column to the table we have

Date	Balance	Days Until Balance Changes
1-May	\$450.21	9
10-May	\$330.21	5
15-May	\$413.64	11
26-May	\$544.10	4
30-May	\$642.33	2

The last entry was 2 since there are 31 days in May.

5. Add a column that contains the balances multiplied by the number of days until the balance changed. We create the

column and multiply the values.

Date	Balance	Days Until Balance Changes	Balance Times Days
1-May	\$450.21	9	\$4,051.89
10-May	\$330.21	5	\$1,651.05
15-May	\$413.64	11	\$4,550.04
26-May	\$544.10	4	\$2,176.40
30-May	\$642.33	2	\$1,284.66

- Find the sum of that last column. Adding that last column we have a sum of \$13,714.04.
- There are 31 days in May, so divide the sum by 31, which gives an average of \$442.39, which is the average daily balance.

> YOUR TURN 6.93

- The billing cycle goes from June 1 to June 30. The previous month's balance is \$563.80. The transactions are in the table below.

Date	Activity	Amount
1-Jun	Balance	\$563.80
2-Jun	Gasoline	\$47.50
2-Jun	Groceries	\$63.42
15-Jun	Movie	\$38.75
15-Jun	Payment	\$250.00
27-Jun	Pharmacy	\$31.21
28-Jun	Auto fuel	\$48.00

Find the average daily balance for this credit card.

Calculating the Interest for a Credit Card

The interest charged for a credit card is based on the daily interest rate of the card, the number of days in the billing cycle, and the average daily balance on the card.

FORMULA

The interest charge, I , for a credit card during a billing cycle is $I = \frac{\text{ADB} \times r \times d}{365}$, where ADB is the average daily balance, r is the annual percentage rate, and d is the number of days in the billing cycle. As before, interest is

rounded up to the next penny.

EXAMPLE 6.94

Calculating Interest for a Credit Card Billing Cycle

Compute the interest charged for the credit card based on the given average daily balance (ABD), annual interest rate, and number of days in the billing cycle.

1. ABD = \$2,765.00, annual interest rate 13.99%, billing cycle of 30 days
2. ABD = \$789.30, annual interest rate 17.99%, billing cycle of 31 days
3. ABD = \$1,037.85, annual interest rate 11.99%, billing cycle of 28 days

Solution

1. Substituting \$2,765.00 for ABD, 0.1399 for r and 30 for d and calculating, we find the interest charge to be

$$I = \frac{\text{ABD} \times r \times d}{365} = \frac{\$2,765.00 \times 0.1399 \times 30}{365} = \$31.80.$$
2. Substituting \$789.30 for ABD, 0.1799 for r and 31 for d and calculating, we find the interest charge to be

$$I = \frac{\text{ABD} \times r \times d}{365} = \frac{\$789.30 \times 0.1799 \times 31}{365} = \$12.06.$$
3. Substituting \$1,037.85 for ABD, 0.1199 for r and 28 for d and calculating, we find the interest charge to be

$$I = \frac{\text{ABD} \times r \times d}{365} = \frac{\$1,037.85 \times 0.1199 \times 28}{365} = \$9.55.$$



YOUR TURN 6.94

Compute the interest charged for the credit card based on the given average daily balance (ABD), annual interest rate, and number of days in the billing cycle.

1. ABD = \$2,135.00, annual interest rate 12.9%, billing cycle of 30 days
2. ABD = \$1,589.63, annual interest rate 9.99%, billing cycle of 31 days
3. ABD = \$6,803.41, annual interest rate 14.9%, billing cycle of 28 days

WHO KNEW?

Credit Cards Charge Stores Fees

The interest you pay is not the only way a credit card company generates revenue. It also charges fees to the retailers, online stores, and service providers that allow the consumer, you, to use your credit card to pay them. These are called processing fees. Currently they typically range from 2.87% to 4.35% of each transaction. That means if you use your credit card at a store and spend \$100.00, the store will have to pay the credit card company somewhere between \$2.87 and \$4.35.

One type of processing fee is the interchange fee. Mastercard charges the vendor 1.35% of the sale, plus an additional percentage up to 3.25%, and a fixed \$0.01 fee for each transaction.

Added to that is an assessment fee. This fee is 0.14% for Visa cards.

Calculating the Balance of a Credit Card

The **balance**, or sometimes balance due, on a credit card is the previous balance, plus all expenses, minus all payments and credits, plus the interest on the card. As stated before, if the card was paid off, there is no interest to be paid.

EXAMPLE 6.95

Calculating the Balance of a Credit Card

Find the balance on the credit card with the given interest charge and balance before interest was charged. The cards

were not paid off previously.

1. Balance before interest is \$708.50, interest charge is \$8.15
2. Balance before interest is \$1,395.10, interest charge is \$21.32

✓ **Solution**

1. Adding the balance before interest to the interest charge, we find the balance to be \$716.65.
2. Adding the balance before interest to the interest charge, we find the balance to be \$1,416.42.

> **YOUR TURN 6.95**

Find the balance on the credit card with the given interest charge and balance before interest was charged. The cards were not paid off previously.

1. Balance before interest is \$560.00, interest charge is \$6.44
2. Balance before interest is \$3,218.00, interest charge is \$49.17

The next example puts all those steps together.

EXAMPLE 6.96

Find Balance Due from Transactions and Interest Rate

Kaylen's credit card charges 16.9% annual interest. His current billing period is from November 1 to November 30. The balance on November 1 was \$1,845.23. Use Kaylen's following transactions to determine his balance due at the end of the billing cycle.

Date	Activity	Amount
1-Nov	Billing Date Balance	\$1,845.23
3-Nov	Groceries	\$78.50
4-Nov	Tablet	\$159.00
4-Nov	Online Game Purchase	\$39.99
4-Nov	Restaurant	\$47.10
10-Nov	Payment	\$300.00
13-Nov	Gasoline	\$58.75
13-Nov	Clothing	\$135.00
18-Nov	Gift	\$30.00
18-Nov	Restaurant	\$21.75
28-Nov	Gasoline	\$43.79

✓ **Solution**

The first step is to find Kaylen's average daily balance. To find the average daily balance, we use the following steps.

1. Start with a list of transactions with their dates and amounts.

This list is provided.

- For each day that had transactions, find the total of the transactions for the day. The days with more than one transaction were Nov. 4, Nov. 13, and Nov. 18. Treating the payment on November 10 as a negative value, the daily transaction amounts are

Date	Amount
1-Nov	\$1,845.23
3-Nov	\$78.50
4-Nov	\$246.09
10-Nov	-\$300.00
13-Nov	\$193.75
18-Nov	\$51.75
28-Nov	\$43.79

- Create a table containing each day with a different balance. The new table with dates that had different balances is below.

Date	Balance
1-Nov	\$1,845.23
3-Nov	\$1,923.73
4-Nov	\$2,169.82
10-Nov	\$1,869.82
13-Nov	\$2,063.57
18-Nov	\$2,115.32
28-Nov	\$2,159.11

- Now, add a column for the number of days those balances until the balance changed. The days until the balance changes is found by finding the difference in the dates. For instance, from May 15 to May 26 was 11. Adding that column to the table we have

The last entry was 3 since there are 30 days in November (the 28th, 29th, and 30th).

Date	Balance	Days Until Balance Changes
1-Nov	\$1,845.23	2
3-Nov	\$1,923.73	1

Date	Balance	Days Until Balance Changes
4-Nov	\$2,169.82	6
10-Nov	\$1,869.82	3
13-Nov	\$2,063.57	5
18-Nov	\$2,115.32	10
28-Nov	\$2,159.11	3

5. Add a column that contains the balances multiplied by the number of days until the balance changed. We create the column and multiply the values.

Date	Balance	Days Until Balance Changes	Balance Times Days
1-Nov	\$1,845.23	2	\$3,690.46
3-Nov	\$1,923.73	1	\$1,923.73
4-Nov	\$2,169.82	6	\$13,018.92
10-Nov	\$1,869.82	3	\$5,609.46
13-Nov	\$2,063.57	5	\$10,317.85
18-Nov	\$2,115.32	10	\$21,153.20
28-Nov	\$2,159.11	3	\$6,477.33

6. Find the sum of that last column. Adding that last column we have a sum of \$62,190.95.
 7. There are 30 days in November, so divide the sum by 30, which gives an average of \$2,073.03, which is the average daily balance.

With the average daily balance, we can determine the interest that is charged for November. Substituting ADB = \$2,073.03, $r = 0.169$, and $d = 30$ into the formula $I = \frac{\text{ADB} \times r \times d}{365}$ and calculating, we find the interest to be

$$I = \frac{\text{ADB} \times r \times d}{365} = \frac{\$2,073.03 \times 0.169 \times 30}{365} = \$28.80.$$

This interest is added to the final balance from the table in step 3, \$2,159.11, which yields a balance due of \$2,101.83.

> YOUR TURN 6.96

1. Angel's credit card charges 16.9% annual interest. His current billing period is from August 1 to August 31. The balance on August 1 was \$982.45. Use Angel's following transactions to determine his balance due at the end of the billing cycle.

Date	Activity	Amount
1-Aug	Billing Date Balance	\$982.45
5-Aug	Food	\$125.31
13-Aug	Payment	\$500.00
14-Aug	Gasoline	\$51.65
14-Aug	Pizza	\$36.99
14-Aug	Shoes	\$89.45
19-Aug	Electric bill	\$178.34
21-Aug	Internet	\$36.99
21-Aug	Food	\$93.45
30-Aug	Gasoline	\$43.18

Minimum Payment Due

The **minimum payment** due is the smallest required amount to be paid on a credit card to avoid late fees and penalties, such as an increased interest rate. The calculations for this may differ from card to card. They also depend in the balance of the credit card. General guidelines for minimum payment due are:

- For larger balances (usually over \$1,000), the minimum payment will be some percentage of the balance due.
- For moderate balances (between \$25 and \$1,000), the minimum would be a specified dollar amount. \$25 seems to be a common value.
- If the balance is small (under \$25 for instance), then the minimum payment is the balance.

Those are just guidelines. Individual cards may vary in these values.

Minimum payments should only be paid if money is short in a given month. The length of time to pay off a credit card using minimum payments is quite long, and results in paying a lot of interest. It is strongly discouraged.

EXAMPLE 6.97

Calculate the Minimum Payment Due

The FYA credit card company has the following minimum payment policy. For balances over \$1,000, the minimum payment is 2.5% of the balance due plus fees, but not interest. For balances between \$500.00 and \$999.99, the minimum payment is \$50.00. For balances \$499.99 and under, the minimum payment is \$25.00 or the balance due, whichever is smaller.

In the following, calculate the minimum payment due given the credit card minimum payment policy, the balance due and fees charged.

1. Balance due is \$1,309.00, no fees
2. Balance due is \$265.50, \$35 in fees
3. Balance due is \$784.90, no fees

Solution

1. The balance is over \$1,000, so the minimum payment is 2.5% of the balance due plus fees. 2.5% of the balance due

is $0.025 \times \$1,309.00 = \32.73 . Since there are no fees, the minimum payment due is \$32.73.

2. The balance is under \$499.99, so the minimum payment due is \$25.00.
3. The balance is between \$500.00 and \$999.00, so the minimum payment due is \$50.00.

> YOUR TURN 6.97

The CLH credit card company has the following minimum payment policy. For balances over \$1,000, the minimum payment is 1% of the balance due plus interest and fees. For balances between \$25.00 and \$999.99, the minimum payment is \$25.00 plus any fees. For balances under \$25.00, the minimum payment is the balance due plus any fees. In the following, calculate the minimum payment due given the credit card minimum payment policy, the balance due and fees charged.

1. Balance due is \$2,308.00, billing cycle interest is \$24.39, no fees
2. Balance due is \$265.50, \$59.00 in fees
3. Balance due is \$19.90, no fees

Check out this [nerdwallet article about minimum payments \(https://openstax.org/r/minimum_payment\)](https://openstax.org/r/minimum_payment) for more!

Check Your Understanding

63. What are the three main types of credit cards?
64. Which type of credit card typically has the highest interest rate?
65. Name three important pieces of information that are included in a credit card statement.
66. A credit card has an average daily balance of \$1,428.50 during a 31-day billing cycle. Find the interest if the interest rate is 15.9%.
67. The billing cycle goes from June 15 to July 14, or 30 days. The balance at the start of the billing cycle is \$3,825.50. The list of transactions on the card is below. Find the average daily balance.

Date	Activity	Amount
15-Jun	Billing Date Balance	\$3,825.50
20-Jun	Food	\$125.31
20-Jun	Clothing	\$345.00
29-Jun	Payment	\$750.00
1-Jul	Restaurant	\$94.80
6-Jul	Gasoline	\$49.75
6-Jul	Home Repair Supplies	\$683.94
6-Jul	Internet	\$49.99
10-Jul	Cell Phone	\$85.00
10-Jul	Food	\$175.24

68. Find the balance due on a credit card statement if ADB = \$487.65, the annual interest rate is 16.8%, the billing cycle was 30 days, and the balance at the end of the billing cycle, before interest is added, was \$689.47.



SECTION 6.10 EXERCISES

1. Name three main criteria to choose a credit card by.
2. Which type of credit card is most convenient to use?
3. Which type of credit card comes with no preset spending limit?
4. How much interest does a credit card holder pay if they pay off the balance every month?
5. Which type of credit card comes with the highest annual fees?

Use the following credit card statement for the following exercises.

September 2022							New Balance \$2,144.89
S	M	T	W	T	F	S	
28	29	30	31	1	2	3	Minimum Payment Due \$50.00
4	5	6	7	8	9	10	
11	12	13	14	15	16	17	Payment Due Date 09/26/22
18	19	20	21	22	23	24	
25	26	27	28	29	30	1	
2	3	4	5	6	7	8	

Late Payment Warning: If we do not receive your minimum payment by the date listed above, you may have to pay a late fee of up to \$39.00.

Minimum Payment Warning: If you make only the minimum payment each period, you will pay more in interest and it will take you longer to pay off your balance. For example:

If you make no additional charges using this card and each month you pay...	You will pay off the balance shown on this statement in about...	And you will end up paying an estimated total of...
Only the minimum payment	8 years	\$3,837
\$76	3 years	\$2,738 (Savings = \$1,099)

Previous Balance	\$1,937.30
Payment, Credits	-\$300.00
Purchases	+\$478.31
Cash Advances	\$0.00
Balance Transfers	\$0.00
Fees Charged	\$0.00
Interest Charged	+\$29.28
New Balance	\$2,144.89
Opening/Closing Date	08/02/22 - 09/01/22
Credit Access Line	\$7,000
Available Credit	\$4,855
Cash Access Line	\$350
Available for Cash	\$350
Past Due Amount	\$0.00
Balance over the Credit Access Line	\$0.00

6. What is the minimum payment due?
 7. How long will it take to pay the balance if the minimum only is paid and no new purchases are made?
 8. How much interest was charged in this billing cycle?
 9. What is the balance on the credit card?
 10. What is the credit limit on this card?
 11. What is the late payment fee for this card?
 12. When is the payment due?
13. The billing cycle for a credit card goes from April 1 to April 30. The balance at the start of the billing cycle is \$1,598.00. The list of transactions on the card is below. Find the average daily balance for the billing cycle.

Date	Activity	Amount
1-Apr	Billing Date Balance	\$1,598.00
9-Apr	Gasoline	\$51.24
9-Apr	Food	\$105.56
9-Apr	Payment	\$675.00
13-Apr	Camping Trip	\$229.75
21-Apr	Gasoline	\$38.45
22-Apr	Gifts	\$148.88
22-Apr	Food	\$49.75
30-Apr	Gym Payment	\$74.99

14. The billing cycle for a credit card goes from September 1 to September 30. The balance at the start of the billing cycle is \$384.25. The list of transactions on the card is below. Find the average daily balance for the billing cycle.

Date	Activity	Amount
1-Sep	Billing Date Balance	\$384.25
2-Sep	Food	\$94.54
5-Sep	Gasoline	\$25.65
5-Sep	Internet	\$39.99
6-Sep	Payment	\$380.00
9-Sep	Insurance	\$174.52
16-Sep	Food	\$83.54
16-Sep	Day Care	\$350.00
20-Sep	Tires	\$2,337.56
21-Sep	Child Clothing	\$27.65
21-Sep	Gasoline	\$31.00
28-Sep	Television	\$299.95

15. The billing cycle for a credit card goes from October 10 to November 9. The balance at the start of the billing cycle is \$930.50. The list of transactions on the card is below. Find the average daily balance for the billing cycle.

Date	Activity	Amount
10-Oct	Billing Date Balance	\$930.50
11-Oct	Clothing	\$350.00
14-Oct	Computer	\$865.84
17-Oct	Food	\$106.51
21-Oct	Payment	\$700.00
21-Oct	Restaurant	\$134.52
21-Oct	Hotel	\$387.56
30-Oct	Hockey Game	\$76.47
5-Nov	Memorabilia	\$150.00
5-Nov	Restaurant	\$94.45
6-Nov	Gasoline	\$49.19

16. The billing cycle for a credit card goes from February 15 to March 16 during a non-leap year. The balance at the start of the billing cycle is \$292.82. The list of transactions on the card is below. Find the average daily balance for the billing cycle.

Date	Activity	Amount
15-Feb	Billing Date Balance	\$292.82
21-Feb	Food	\$64.57
22-Feb	Gasoline	\$31.50
22-Feb	Food	\$71.94
28-Feb	Insurance	\$133.25
28-Feb	Payment	\$100.00
3-Mar	Gasoline	\$26.61
12-Mar	School Trip	\$300.00

In the following exercises, compute the interest charged for the credit card based on the given average daily balance (ADB), annual interest rate, and number of days in the billing cycle.

17. ADB = \$350.00, annual interest rate 14.9%, billing cycle of 30 days.
18. ADB = \$4,312.00, annual interest rate 9.99%, billing cycle of 31 days.
19. ADB = \$563.38, annual interest rate 17.9%, billing cycle of 30 days.
20. ADB = \$1,043.53, annual interest rate 11.9%, billing cycle of 31 days.

In the following exercises, find the balance on the credit card with the given interest charge and balance before interest was charged. The cards were not paid off previously.

21. Balance before interest is \$1630.00, interest charge is \$16.48.
 22. Balance before interest is \$621.00, interest charge is \$7.81.
 23. Balance before interest is \$1,380.00, interest charge is \$15.35
 24. Balance before interest is \$2,774.00, interest charge is \$44.05.
25. Alsaggr's credit card charges 11.9% annual interest. His current billing period is from April 1 to April 30. The balance on April 1 was \$1,598.00. Use Alsaggr's following transactions to determine his balance due at the end of the billing cycle.

Date	Activity	Amount
1-Apr	Billing Date Balance	\$1,598.00
9-Apr	Gasoline	\$51.24
9-Apr	Food	\$105.56
9-Apr	Payment	\$675.00
13-Apr	Camping Trip	\$229.75
21-Apr	Gasoline	\$38.45
22-Apr	Gifts	\$148.88
22-Apr	Food	\$49.75
30-Apr	Gym Payment	\$74.99

26. Marisa's credit card charges 8.9% annual interest. Her current billing period is from September 1 to September 30. The balance on September 1 was \$384.25. Use Marisa's following transactions to determine her balance due at the end of the billing cycle.

Date	Activity	Amount
1-Sep	Billing Date Balance	\$384.25
2-Sep	Food	\$94.54
5-Sep	Gasoline	\$25.65
5-Sep	Internet	\$39.99
9-Sep	Insurance	\$174.52
16-Sep	Food	\$83.54
16-Sep	Day Care	\$350.00
20-Sep	Tires	\$2,337.56
21-Sep	Child Clothing	\$27.65

Date	Activity	Amount
21-Sep	Gasoline	\$31.00
28-Sep	Television	\$299.95

27. Haley's credit card charges 18.9% annual interest. Her current billing period is from October 10 to November 9. The balance on October 10 was \$930.50. Use Haley's following transactions to determine her balance due at the end of the billing cycle.

Date	Activity	Amount
10-Oct	Billing Date Balance	\$930.50
11-Oct	Clothing	\$350.00
14-Oct	Computer	\$865.84
17-Oct	Food	\$106.51
21-Oct	Payment	\$700.00
21-Oct	Restaurant	\$134.52
21-Oct	Hotel	\$387.56
30-Oct	Hockey Game	\$76.47
5-Nov	Memorabilia	\$150.00
5-Nov	Restaurant	\$94.45
6-Nov	Gasoline	\$49.19

28. Pavly's credit card charges 10.9% annual interest. His current billing period is from February 15 to March 16 in a non-leap year. The balance on February 15 was \$292.82. Use Pavly's following transactions to determine his balance due at the end of the billing cycle.

Date	Activity	Amount
15-Feb	Billing Date Balance	\$292.82
21-Feb	Food	\$64.57
22-Feb	Gasoline	\$31.50
22-Feb	Food	\$71.94
28-Feb	Insurance	\$133.25
28-Feb	Payment	\$100.00